

HAT-P-54b

R-1247

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_211222 · created 2026-07-31T11:31:34+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459570.9721 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.199 +/- 0.059
Transit depth (Rp/Rs)^2	3.96 +/- 2.35 [%]
Orbital Inclination (inc)	89.56 +/- 2.2
Airmass coefficient 1 (a1)	2.12 +/- 0.52
Airmass coefficient 2 (a2)	-0.57 +/- 0.18
Scatter in the residuals of the lightcurve fit is	23.05 %
Best Comparison Star	#1 - [624, 46]
Optimal Aperture	1.65
Optimal Annulus	7.05
Transit Duration (day)	0.0727 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.199 ± 0.059 vs 0.1572 (deviation 0.37σ)
Duration fit vs published	0.0727 d vs 0.075 d (deviation 0.2σ)
Mid-time O–C	-8.9 min
Depth SNR	1.8
Residual scatter	23.05 %

Light curve

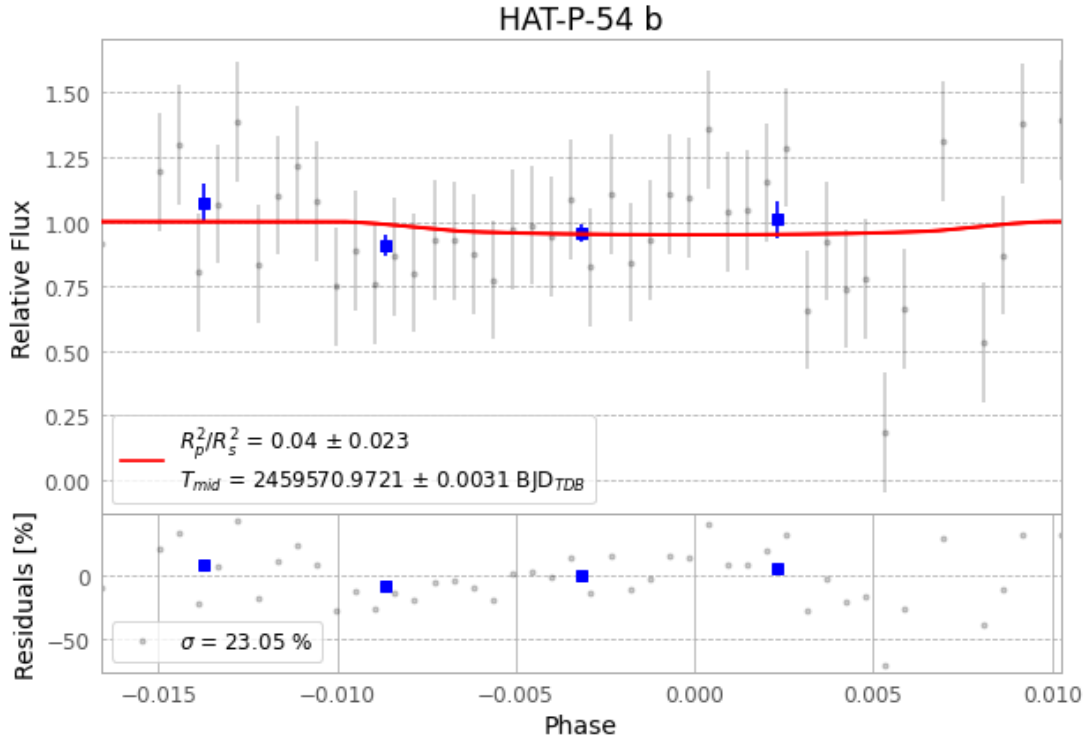


Figure 1 — FinalLightCurve_HAT-P-54 b_2021-12-22.png (SHA-256 06c80aa5fd3a965f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [316, 273], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6ae8c2ba68b5b1fa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

50 FITS frames (33 MB), HATP-54211222093912.FITS → HATP-54211222120612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-211222.log (83b7e0f1bb57...), FinalLightCurve_HAT-P-54 b_2021-12-22.png (06c80aa5fd3a...), FinalLightCurve_HAT-P-54 b_2021-12-22.csv (111fea72764f...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 11:31Z.